

AI Operations (AIOps)

Module 1 Assignment — Experiment Management & Reproducibility

Total Marks	Weight	Submission
50	10% of course grade	Git repository link + a 1-page PDF write-up

Question 1 — Conceptual: Technical Debt Diagnosis (10 marks)

A campus food-delivery startup has a recommendation model in production. Over the last two months, engineers report:

(a) changing the “estimated delivery time” feature's rounding logic unexpectedly hurt recommendations for a completely unrelated “favorite restaurants” feature;

(b) a marketing dashboard team has been silently reading the model's raw output table for a campaign and nobody on the ML team knew;

(c) the training pipeline is a chain of 14 undocumented shell scripts with no orchestration tool.

1. Identify which of the six hidden-technical-debt categories (from Lecture 1) best explains each of (a), (b), and (c). (6 marks)
2. For ONE of the three, propose a specific mitigation using a tool or practice introduced in the lectures covered so far. (4 marks)

Question 2 — Applied: MLflow Experiment Comparison (15 marks)

Using the training script provided in the Lecture 3 exercises, run at least SIX experiments varying at least two hyperparameters (e.g., learning rate and batch size, or architecture), after changing the predictor from RandomForest to an MLP. Also, change the dataset from IRIS to MNIST.

Submit:

1. A screenshot of the MLflow run-comparison table for all six runs. (4 marks)
2. A short written analysis (150–250 words) identifying: the best-performing run and why; any evidence of overfitting from the train_loss vs. val_accuracy trend; and which hyperparameter appears to have the larger effect on performance. (7 marks)
3. The exact mlflow.log_param / mlflow.log_metric code you added to the starter script. (4 marks)

Question 3 — Applied: DVC Data Versioning & Rollback (10 marks)

Starting from the provided dataset (data.zip used in the class), let's create a CSV of the file names, which will have 1800 rows + a header line.

1. Initialize DVC in a Git repo and push (let's call it) v1 to a remote (only SSH remote or any S3 bucket are acceptable). (2 marks)
2. Simulate a data update: add new rows from new_labels.zip to create v2 (now, the row count should 2801), re-run dvc add, and commit v2 with a clear commit message. (3 marks)
3. Demonstrate rolling back to v1 using git checkout + dvc checkout, and include terminal output (screenshot or copy-pasted) proving the row count matches v1 exactly. (5 marks)

Question 4 — Capstone: End-to-End Reproducibility Drill (15 marks)

Consecutive roll numbers will become partners for this experiment. Working with an assigned partner, complete the full reproducibility protocol:

1. Partner A trains a model, logs the run in MLflow (params, metrics, seed, git_commit tag, artifact), versions the dataset with DVC, and commits code + .dvc file in the same Git commit. Registers the model and transitions it to “Staging.” (6 marks)
2. Partner A shares the repository with Partner B (no other communication about environment/data allowed).
3. Partner B reproduces the result using only: git clone, git checkout <commit>, dvc checkout, mamba* env create -f environment.yml, and reruns the script. (6 marks) *[mamba or conda or virtualenv]
4. Partner B logs a note in the MLflow run documenting whether the metric matched (within a stated tolerance) or explaining any discrepancy. (3 marks)

Overall Rubric Summary

Question	Marks	Skill Assessed
Q1	10	Conceptual understanding of ML technical debt
Q2	15	Practical MLflow experiment tracking & analysis
Q3	10	Practical DVC data versioning & rollback
Q4	15	Integrated end-to-end reproducibility (capstone)
Total	50	